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What is Ollama
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** Ollama is a tool used to run Large Language Models in our local computers.

** Usually, when we use OpenAI API / Google Models, our question goes to a cloud server and it works based on subscription (we need to pay money for using them).

** Ollama lets you run LLMs in our machine (free of cost).

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What can we do with Ollama?
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using Ollama, we can:

** Run AI models in our laptop.

** Build a local chatbot.

** Use LLMs without any API key.

** Test Generative AI projects in offline mode.

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Popular models in ollama
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- 1) llama3.2 : General chatbot
- 2) mistral : Text generation and reasoning
- 3) gemma : light weight model from Google
- 4) qwen : Coding related task
- 5) deepseek-r1 : Reasoning based tasks
- 6) codellama : Code generation

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Advantages with ollama
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** Easy to install

** Simple commands

** Runs model locally

** Good for learning purpose

** No API key required

** Free of cost

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Limitations of Ollama

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** Large models need high RAM or GPU

** Small models may give weaker answers than OPEN GPT Models

** Response speed depends on system configuration.

Note : Minimum 8 GB RAM is required (16 GB RAM is better)

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OLLAMA Setup in our machine

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Step-1 : Download and install ollama software (.exe file)

URL : <https://ollama.com/download/windows>

Step-2 : After installation, Open "command prompt" and verify ollama version

\$ ollama --version

show models which are already downloaded

\$ ollama list

download a model based on its name

\$ ollama pull <model-name>

display model summary

\$ ollama show <model-name>

execute llm in local machine

\$ ollama run <model-name>

display models which are running in our machine

\$ ollama ps

Stop the model execution

\$ ollama stop <mode-name>

copy the model

\$ ollama cp llama3.2 my-llama-model

Remove model from our machine

\$ ollama rm <model-name>

Note-1: Expose our Local Model as API using ollama software for integration in our gen ai project.

\$ ollama serve

Note: After executing above command, verify ollama api is running or not

URL : <http://localhost:11434/>

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GEN AI Project Development using OPEN AI GPT Model

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Step-1 : Create project folder using IDE

Ex : 02-ollama-gpt-chat-bot

Step-2 : Create Python Virtual Environment and activate it using below commands

\$ cd 01-openai-gpt-chat-bot

\$ python -m venv venv

\$ venv\Scripts\activate

Step-3 : Create "requirements.txt" and configure libraries and install them using pip

```
```
ollama
streamlit
```
```

\$ pip install -r requirements.txt

Step-5 : Create "backend.py" to consume OLLAMA based LLM

```
input : model, prompt
output : LLM Response
```

Step-6 : Create frontend.py with streamlit ui (like chatgpt ui)

Step-7 : Run the application and test it.

\$ streamlit run frontend.py

===== backend.py =====

```
import ollama
```

```
MODEL_NAME = "llama3.2:latest"
```

```
def ask_ollama(question):
    response = ollama.chat(
        model=MODEL_NAME,
        messages=[
            {
                "role": "user",
                "content": question
            }
        ]
    )
    return response["message"]["content"]
```

===== frontend.py =====

```
import streamlit as st
from backend import ask_ollama
```

```
# =====
# Page Configuration
# =====
```

```
st.set_page_config(
    page_title="Ollama Chatbot",
    page_icon="ðŸ’–",
    layout="centered"
)

# =====
# Page Header
# =====

st.title("ðŸ’– Ollama Chatbot")
st.caption("Powered by Llama 3.2")

# =====
# Chat Input
# =====

question = st.chat_input("Ask me anything...")

# =====
# Process Question
# =====

if question:

    # User message
    with st.chat_message("user"):
        st.write(question)

    # Assistant message
    with st.chat_message("assistant"):

        with st.spinner("Thinking..."):

            answer = ask_ollama(question)

            st.write(answer)

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```